

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

Anonymous authors
Anonymous submission

ABSTRACT

Vision–language models (VLMs) are attractive candidates for zero-shot goal-conditioned value functions, yet their frozen pre-trained representations curtail precisely the two capabilities on which reliable value estimation depends: generalization to unseen conditions and temporal reasoning over observation histories. We introduce VITA, a zero-shot value function estimator that resolves both limitations through test-time adaptation, dispensing with the additional pre-training and task-specific supervision on which prior methods rely. At inference, a lightweight adaptation module is updated by a single gradient step on a meta-learned self-supervised loss whose objective is shaped during training so that each update measurably improves value estimation. Because these updates are applied sequentially along a trajectory, VITA encodes the observation history directly into the module’s parameters, thereby furnishing the temporal reasoning that frozen contrastive representations otherwise lack. To preclude shortcut learning, we introduce a dissimilarity-based sampling strategy that concentrates training on the most semantically diverse sub-trajectories. Trained on a deliberately narrow slice of real-world robotic manipulation—pick-and-place demonstrations collected with a single embodiment in one environment—VITA nonetheless generalizes to a broad range of out-of-distribution tasks, environments, and embodiments, consistently outperforming a state-of-the-art zero-shot estimator built on autoregressive VLMs. We further show that VITA’s zero-shot value estimates constitute an effective reward-shaping signal for offline reinforcement learning, yielding multi-task policies on the Meta-World MT10 benchmark whose interquartile-mean return exceeds that of policies trained with the simulator’s hand-engineered fuzzy-logic dense reward.

1 INTRODUCTION

Vision–language models (VLMs) exhibit strong generalization across diverse tasks and domains, a capability acquired by learning from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). Sequential decision-making has yet to attain comparable generality: despite substantial progress in learning generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), prevailing methods remain bottlenecked by their reliance on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer a compelling alternative, allowing agents to learn within simulated environments across applications that span robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even as world models render increasingly realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), a decisive obstacle persists: *how can agents learn from video at scale without relying on human supervision?*

One line of research learns policies directly from expert visual trajectories by inferring the actions that connect successive visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a); the resulting latent actions can in principle be mapped to executable controls at deployment, although grounding them reliably remains difficult in practice. A complementary and increasingly influential direction instead learns a universal *goal-conditioned value function* that supplies a zero-shot reward-shaping signal for reinforcement learning (RL) and a supervision signal

for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this framing, goal-conditioned value estimation reduces to a well-posed prediction problem: estimating how far an agent has progressed toward completing a task, conditioned jointly on its visual observations and a natural-language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025). The formulation is attractive precisely because it decouples supervision from action labels, thereby opening a path to learning from video at scale.

Prior work has repurposed pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping, exploiting the similarity between a task description and each visual observation within a shared multimodal representation space to ground task progress in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Such frame-level similarity, however, carries no temporal context and therefore cannot disambiguate visually similar states that recur at different stages of a task—folding versus unfolding a shirt being the canonical example. Autoregressive VLMs (Alayrac et al., 2022) instead inject temporal context by conditioning on the entire visual trajectory within the prompt, but they inherit a systematic bias toward monotonically increasing predictions from the chronologically ordered corpora on which they are pre-trained (Ma et al., 2024). The two architectures thus fail for complementary reasons while sharing a single root cause: their reliance on frozen pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) for zero-shot prediction simultaneously bounds both their generalization and their temporal reasoning (Pătrăucean et al., 2023). Attempts to recover generalizable value functions have consequently resorted to large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demonstration-based reward specification (Sontakke et al., 2023)—each of which reintroduces precisely the dependence on scale or supervision that zero-shot estimation is meant to avoid.

We introduce VITA, a zero-shot value function estimator that resolves both limitations at once by adapting at test time, and that outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation. Rather than leaning on large-scale pre-training or task-specific demonstrations to generalize, VITA adapts online to the semantic and temporal context of each trajectory. At inference, a lightweight adaptation module is updated in negligible time by a single gradient step on a meta-learned self-supervised loss (Sun et al., 2024) whose objective is shaped during training so that every test-time update measurably improves value estimation (Finn et al., 2017). Applying these updates sequentially over a test trajectory drives the estimator to encode the trajectory history directly into its parameters (Sun et al., 2024); this accumulation is precisely the mechanism through which VITA supplies the temporal reasoning that prior zero-shot methods lack. To preclude shortcut learning (Geirhos et al., 2020), we further introduce a dissimilarity-based sampling strategy that steers the estimator toward genuine semantic cues, an effect we substantiate empirically. Our evaluation is organized around the three capabilities that jointly characterize an effective value function estimator—generalization under distribution shifts, discrimination between expert and non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020)—and VITA delivers on all three.

Our contributions are as follows:

- **Method.** We propose VITA, a meta-learned test-time adaptation method that endows contrastive VLMs with the generalization and temporal reasoning required for zero-shot value function estimation, encoding trajectory history through sequential per-timestep parameter updates rather than through frozen representations, and requiring neither task-specific demonstrations nor large-scale pre-training.
- **Generalization.** We show that, trained on pick-and-place demonstrations from a single environment and embodiment, VITA generalizes to a broad range of out-of-distribution tasks, environments, and embodiments in robotic manipulation, attaining the best or second-best value order correlation on every evaluation dataset and consistently outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- **Downstream utility.** We demonstrate that VITA’s zero-shot value estimates serve as an effective reward-shaping signal, producing offline RL policies on the Meta-World multi-task benchmark (MT10) whose interquartile-mean return surpasses that of policies trained with the simulator’s hand-engineered fuzzy-logic dense reward (Yu et al., 2020)—even though VITA is trained exclusively on real-world robot video.

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task progress estimation is naturally cast as goal-conditioned value function estimation in which the reward encodes task completion. We accordingly formulate the learning problem as predicting the degree of task completion from vision–language representations. Formally, the value function $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar quantifying the predicted progress toward goal completion, with the endpoints fixed by convention: $V(o_t; g) = 0$ denotes the onset of the task and $V(o_t; g) = 1$ its completion. Following common practice, we identify task progress with temporal position in expert demonstrations, invoking the standard assumption that expert trajectories advance monotonically toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Concretely, given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined through normalized timestep indices, $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T is the trajectory length. Normalized temporal position thereby furnishes the supervisory signal for learning the goal-conditioned value function V , a proxy that is well-founded for the monotonic expert demonstrations considered here.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is a test-time adaptation paradigm that recasts inference itself as a self-supervised learning problem (Goodfellow et al., 2016), updating model parameters on each test instance without access to labels. Although originally formulated for static image classification (Sun et al., 2020), TTT has since been extended to sequence modeling (Sun et al., 2024; Wang et al., 2023), in which a parametric adaptation module f_{adapt} is updated at every timestep by a gradient step on a self-supervised loss. Whereas conventional sequence architectures retain temporal history in activation states—recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key–value pairs (Vaswani et al., 2017)—TTT instead encodes that history into the parameters of f_{adapt} through sequential updates performed during inference, so that the parameters themselves function as an implicit memory that accumulates temporal history while preserving temporal order. Sun et al. (2024) strengthen this scheme by meta-learning the self-supervised task under the gradient-based meta-learning paradigm (Finn et al., 2017): in this formulation (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and optimized to minimize the downstream prediction loss *after* a test-time update, rather than being minimized directly. We build on this paradigm and adapt it to goal-conditioned value function learning, in which temporal context is captured through sequential parameter updates and the self-supervised task is meta-learned to improve value estimation rather than fixed a priori.

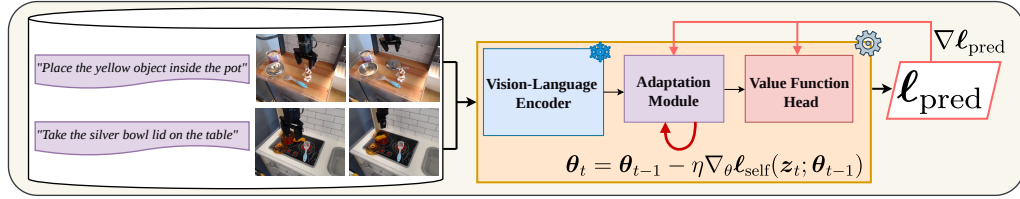
3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

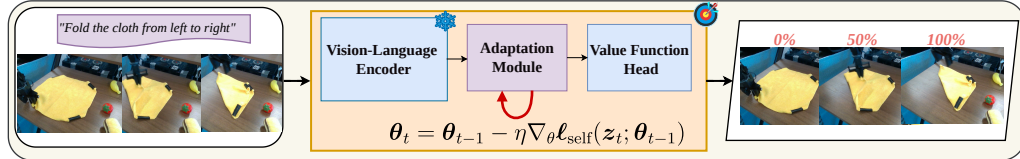
Our goal-conditioned value function estimator comprises three tightly coupled modules: (1) a frozen multimodal encoder that extracts joint vision–language representations from visual trajectories and their task descriptions; (2) an adaptation module whose parameters are updated at test time by a self-supervised loss that is itself meta-learned to improve value estimation; and (3) a regression head that maps the adapted representation to a scalar value estimate. The architecture deliberately confines all test-time adaptation to the second module: generalization and temporal reasoning are therefore acquired online, at inference, while the encoder and regression head remain fixed. We detail each module in turn below.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We employ a frozen contrastive vision–language encoder, CLIP (Radford et al., 2021), to map visual observations and goal descriptions into a shared representation space. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate the corresponding embeddings at each timestep t into a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$ and ϕ_v and ϕ_g denote the visual and language encoders, respectively.



(a) Training: meta-learning the goal-conditioned value function estimator so that a single test-time update improves progress prediction.



(b) Inference: producing a zero-shot value estimate for an out-of-distribution test trajectory via sequential test-time adaptation.

Figure 1: **Overview of VITA.** VITA meta-learns a goal-conditioned value function during training and, at inference, adapts it online to each out-of-distribution trajectory through sequential test-time adaptation, thereby achieving zero-shot generalization without task-specific demonstrations.

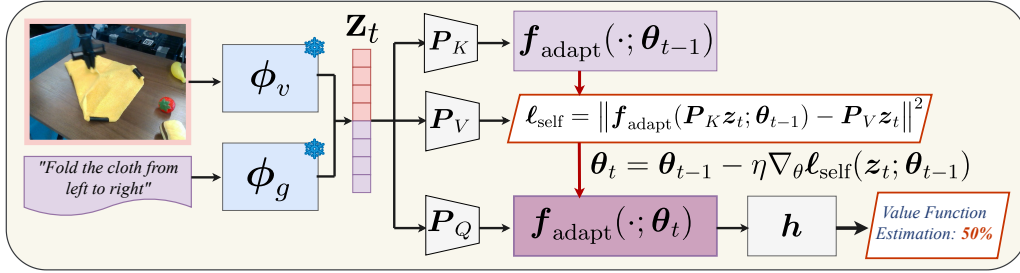


Figure 2: **Test-time adaptation.** At inference, at each timestep t the adaptation module f_{adapt} is updated by a single gradient step on the meta-learned self-supervised loss ℓ_{self} . Applied sequentially along the trajectory, these updates accumulate the observation history in the module’s parameters, yielding an implicit temporal memory that supplies temporal reasoning without any explicit recurrent state.

Because CLIP is pre-trained with a contrastive objective that aligns paired image–text inputs within this shared space (Radford et al., 2021), observations semantically closer to goal completion lie nearer the goal-description embedding; VITA exploits this alignment directly as the semantic substrate on which value estimation is built.

3.1.2 TEST-TIME ADAPTATION

To adapt the multimodal representations to both semantic and temporal context, we introduce an adaptation module f_{adapt} instantiated under the test-time training paradigm of Section 2.2. At inference, the parameters of f_{adapt} are updated online by a gradient step on a meta-learned self-supervised loss ℓ_{self} . We formulate this loss as a reconstruction objective mediated by two learnable linear projections P_K and P_V , both meta-learned during training so that each adaptation step improves value estimation. Concretely, with $P_K \in \mathbb{R}^{d' \times 2d}$ producing a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ producing the reconstruction target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Through these sequential updates, f_{adapt} implicitly retains information from past visual observations and so captures temporal context without any explicit recurrent state. Section 4.6.2 provides direct empirical evidence that incremental, per-timestep parameter updates yield substantially more accurate value estimation than updates aggregated at the window or trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into the adaptation space $\mathbb{R}^{d'}$. The projected representation is processed by the adapted module f_{adapt} and passed to a regression head h , which emits the predicted value of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), mirroring the value-function estimator architectures common in deep reinforcement learning (Espeholt et al., 2018). It is trained against the normalized progress labels y_t derived from expert demonstrations (Section 2.1) under a supervised prediction loss ℓ_{pred} , the mean squared error between the predicted values $V(z_t; g)$ and the targets y_t . Crucially, at inference the regression head h is held frozen while only the adaptation module f_{adapt} is updated online through the self-supervised objective. This asymmetry—a fixed prediction head atop an online-adapted representation—is precisely what enables the estimator to generalize zero-shot to unseen trajectories without any task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator with gradient-based meta-learning, shaping the self-supervised loss ℓ_{self} so that a single test-time adaptation step reduces the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training proceeds over diverse sub-trajectories selected by dissimilarity-based sampling, a procedure that suppresses shortcut learning and redirects the estimator toward genuine semantic and temporal cues. We describe the two components—meta-learning and dissimilarity-based sampling—in turn.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep by the self-supervised loss ℓ_{self} . Following the gradient-based meta-learning paradigm (Finn et al., 2017), we backpropagate through these inner adaptation updates with respect to the supervised prediction loss ℓ_{pred} , jointly optimizing the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . The total training objective combines the supervised prediction loss ℓ_{pred} with the self-supervised loss ℓ_{self} , weighted by a scalar λ ; the optimized θ_0 subsequently serves as the initialization for adaptation at inference. The estimator therefore does not merely learn to predict progress: it learns *how to adapt* its own internal representations at test time so as to improve value estimation on trajectories it has never encountered.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories are typically dominated by highly redundant consecutive frames, a phenomenon well documented in video action recognition (Wang et al., 2018). Left unaddressed, this redundancy invites the value function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b)—overfitting to frequently recurring late-stage visual patterns rather than to task semantics. We counter this failure mode with a dissimilarity-based sampling strategy that assembles each mini-batch from the most visually diverse sub-trajectories within a trajectory. By deliberately inflating intra-batch variance, the strategy operates as a form of importance sampling that upweights the semantically informative segments the estimator would otherwise neglect.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories with a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. The

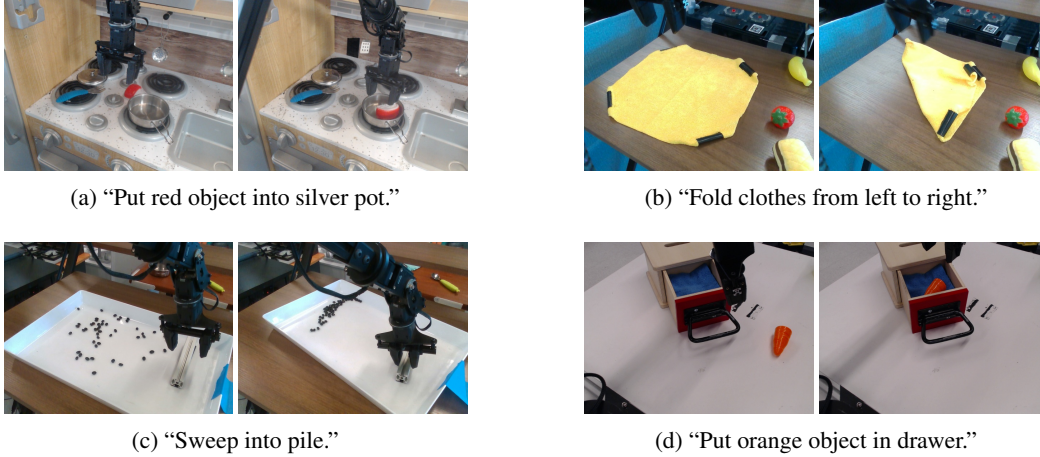


Figure 3: Representative visual trajectories paired with their task descriptions, illustrating the distribution shifts under which VITA is evaluated. Each shift departs progressively further from the pick-and-place, single-embodiment training regime. (a) In-distribution. (b, c) Environment shift. (d) Joint embodiment and environment shift.

maximally diverse subset of size k is the one maximizing total pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Exact optimization of Eq. 4 is combinatorially intractable, since it requires enumerating all $\binom{|\mathcal{W}|}{k}$ subsets. We therefore approximate the objective with an efficient scoring-based heuristic that assigns each window $w \in \mathcal{W}$ a diversity score equal to its aggregate dissimilarity from all other windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Scoring every window by s and retaining the k highest-scoring ones thus recovers a diverse subset in polynomial time. As the complexity analysis in Appendix G confirms, the resulting overhead is negligible relative to the cost of model training. Our ablation study (Section 4.6.1) validates the design empirically: dissimilarity-based sampling markedly improves the estimator’s ability to distinguish expert from non-expert visual trajectories relative to full-trajectory sampling.

4 EXPERIMENTS

Our evaluation is organized around the three capabilities that jointly characterize an effective goal-conditioned value function, and it establishes that VITA satisfies each: (i) generalization of value estimation for real-world robotic manipulation under distribution shifts in task, environment, and embodiment, including their combination; (ii) reliable discrimination between expert and non-expert visual trajectories in real-world robotic manipulation; and (iii) effective zero-shot reward shaping for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark. These three axes probe generalization, robustness, and downstream utility, respectively.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions drawn from BridgeData V2 (Walke et al., 2023), a dataset spanning a wide range of manipulation tasks, environments, and robot embodiments. Our training subset is deliberately narrow: it comprises 2,986 expert demonstrations of pick-and-place manipulation alone—no folding, sweeping, or stacking—all collected with a single robot embodiment (WidowX 250) across four configurations of the ToyKitchen

environment. This restricted regime renders the out-of-distribution generalization reported below a stringent test rather than an incidental byproduct of broad training coverage. A further 287 expert demonstrations are held out as the in-distribution test set, denoted `tk_pnp`; representative training examples appear in Appendix A. Both video frames and task descriptions are encoded with a frozen OpenCLIP ViT-B/32 backbone. The training objective combines ℓ_{pred} with ℓ_{self} at weight $\lambda_{\text{self}} = 0.5$, and dissimilarity-based sampling uses window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, a single gradient step ($t_{\text{ep}} = 1$) on ℓ_{self} with learning rate $\eta = 0.1$ suffices for adaptation. Complete details and hyperparameter sweeps are reported in Appendix D.

4.2 BASELINES

We benchmark VITA against a representative set of value functions that spans both the contrastive and the autoregressive VLM families. VLM-CL derives zero-shot value estimates from the cosine similarity between the CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, whereas CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same representations to model temporal history through recurrent hidden states—furnishing the closest recurrent counterpart to VITA’s implicit memory. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function built on autoregressive VLMs; we evaluate it in two regimes: GVL-0S, the zero-shot setting in which the VLM is prompted with only the test trajectory and task description, and GVL-1S, a one-shot in-context setting that additionally supplies a full shuffled trajectory from the training distribution together with its progress labels. Following GVL, we use Gemini 1.5 Pro (`gemini-1.5 Pro-latest`) (Gemini Team et al., 2024) with the prompt template proposed therein. We additionally probed two alternative backbones: Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome the monotonic temporal bias, and GPT-4o (Hurst et al., 2024), which declined to emit scalar value estimates. Additional implementation details, including the parameter and tuning budgets granted to each baseline, appear in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: Value Order Correlation (VOC) for zero-shot value function estimation under distribution shift, evaluated per dataset; higher is better. **Bold** marks the best method and underline the second best in each row. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts. Across all ten datasets, VITA and the recurrent CLIP-GRU baseline dominate both the autoregressive GVL variants and the temporally unaware CLIP baselines; VITA attains the best or second-best VOC on every dataset and the best on 6 of 10.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

We assess how well progress estimators generalize across novel environments, tasks, and robot embodiments using subsets of BridgeData V2 curated to induce controlled variation along each axis. Environment shifts alter scene layout or background: `lm_pnp` is a pick-and-place task performed in front of a laundry machine, `td_fold`, `ft_fold`, and `rd_fold` involve cloth folding on distinct surfaces, and `ms_sweep` introduces a long-horizon sweeping task within a confined tray.

Embodiment shifts are induced with the DeepThought robot, which differs from the training embodiment (WidowX 250) in both morphology and camera perspective: `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) hold the in-distribution environment fixed under the new embodiment, whereas `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) compound embodiment and environment shifts simultaneously. Each dataset comprises 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 owing to its limited size; Appendix B describes the evaluation datasets in full. Performance is quantified by the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which measures the agreement between predicted progress toward task completion and the chronological ordering of the frames in a trajectory. Formally, letting $p_1, \dots, p_T$ denote the predicted progress values for a trajectory of T frames and $r_t = t$ the temporal index, VOC is the Spearman rank correlation ρ_s between the predicted values and the temporal indices. As a rank-order statistic, VOC measures monotonic agreement with expert progression rather than the absolute calibration of the value estimates.

Table 1 establishes a clear ordering among the methods. Both VITA and CLIP-GRU outperform GVL-0S and GVL-1S across the board, exposing the limitations of pre-trained autoregressive VLMs and underscoring the importance of preserving temporal order for progress estimation. Between the two temporal models, VITA is the stronger: it attains the best or second-best VOC on every one of the ten datasets and the best on 6 of 10, surpassing CLIP-GRU in the majority of settings. Both temporal models decisively outperform the CLIP-based baselines that omit temporal modeling (CLIP-FT, VLM-CL, VLM-RM)—a gap that quantifies the value of encoding temporal history at all. CLIP-FT is competitive with GVL in-distribution (0.251 vs. 0.269 on `tk_pnp`) but collapses out-of-distribution, while VLM-CL and VLM-RM perform poorly throughout—indeed near or below zero on `ms_sweep`—consistent with their absence of temporal modeling. The in-context examples supplied to GVL-1S help on some datasets yet prove neutral or even detrimental on others, indicating that few-shot prompting is an unreliable route to generalization. Tellingly, both GVL variants perform well on folding (`td_fold`, `ft_fold`, `rd_fold`) yet degrade sharply on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`), suggesting that the underlying autoregressive VLM is biased toward folding-like manipulations. VITA, by contrast, sustains consistent accuracy across every task type and distribution shift, evidencing markedly stronger generalization than in-context learning.

On the long-horizon `ms_sweep` task, every method except CLIP-FT degrades, yet VITA still attains the highest VOC (0.490) and, notably, surpasses CLIP-GRU (0.434)—direct evidence that, for long-horizon manipulation, encoding temporal history through test-time adaptation is more effective than storing it in recurrent hidden states. VITA generalizes comparably well under embodiment shift. On `dt_tk_pnp`, which pairs a new robot embodiment with the training environment and tasks, both VITA (0.820) and CLIP-GRU (0.856) exceed their in-distribution scores, indicating that learned temporal context transfers across embodiments rather than merely memorizing the training morphology. GVL-1S, by contrast, performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, reinforcing that few-shot in-context learning does not generalize under embodiment shift in our evaluation setting.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, a dependable value function must be robust: it should assign lower progress to suboptimal behavior than to expert behavior. We therefore test VITA’s capacity to separate expert from non-expert trajectories by comparing its predictions on expert and scripted trajectories drawn from the same in-distribution setting of BridgeData V2. The scripted trajectories are generated in the ToyKitchen environment by a heavily randomized controller (Walke et al., 2023), under the same embodiment and environmental configuration as the expert demonstrations, and span pick-and-place tasks over object categories that may overlap with the training set: general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution in task, embodiment, and environment, their execution departs from the smooth, monotonic progress characteristic of expert demonstrations; Appendix C provides examples. Because the randomized controller does not target calibrated levels of suboptimality, the scripted data carry binary expert/non-expert labels rather than quantifiable degrees of failure; even so, an effective estimator evaluated in this matched setting should assign systematically lower VOC to the suboptimal trajectories than to the expert ones. We capture this with *BinVOC*, a binary metric defined as $1[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which

equals 1 exactly when expert trajectories receive higher VOC than their suboptimal counterparts and 0 otherwise.

Discriminative performance is reported as BINVOC averaged over the five scripted datasets, with expert performance (VOC_{exp}) measured on the in-distribution test set `tk_pnp`, whose pick-and-place tasks parallel those in the scripted evaluation. As Table 2(a) reports, VITA, GVL-0S, and GVL-1S achieve perfect discrimination ($\text{BINVOC} = 1.00$), assigning higher VOC to expert trajectories on every task. Critically, VITA outperforms CLIP-GRU (0.80) here, indicating that storing temporal memory in recurrent hidden states is more prone to overfitting temporal shortcuts than VITA’s implicit memory formed through sequential test-time updates. The CLIP-based baselines VLM-CL (0.40) and VLM-RM (0.00) underperform, consistent with their absence of temporal modeling, and CLIP-FT (0.80), though more reliable, still fails on one task—evidence of sensitivity to object distribution shift. VITA therefore unites the perfect discrimination of the autoregressive VLMs with the strong out-of-distribution generalization the GVL variants lack.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

To probe downstream utility, we evaluate the estimator on the Meta-World MT10 benchmark (Yu et al., 2020), comprising ten diverse robotic manipulation tasks that span pick-and-place, door opening, and pushing. The experiment is deliberately demanding: VITA is trained solely on real-world robotic video and applied zero-shot to this simulated domain, bridging a substantial real-to-sim gap—the reverse of the conventional sim-to-real transfer direction. For each MT10 task we generate 20 expert visual demonstrations using Meta-World’s expert policies, and the goal-conditioned estimator supplies a dense reward for every visual observation. Policies are then trained with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL regime. We repeat expert data generation and policy training over 10 random seeds; each resulting multi-task policy is evaluated on 20 rollouts per task, and we compute the average return per (task, seed) pair. Following the protocol of Agarwal et al. (2021), we report the interquartile mean (IQM) across all pairs together with 95% stratified bootstrap confidence intervals, so that reported differences are accompanied by a characterization of their uncertainty. Full training and evaluation details are given in Appendix E.

Table 2(b) reports MT10 performance under this protocol. VITA attains the highest IQM return (0.815, 95% CI [0.785, 0.838]), ranking above every CLIP-based baseline; its confidence interval lies entirely above the point estimates of the learned reward baselines, though it partially overlaps their intervals, so we read the gain as a consistent improvement rather than a categorical separation. A direct comparison against GVL is infeasible at this scale: GVL relies on Gemini-1.5, a proprietary autoregressive VLM whose per-observation inference cost is prohibitive in RL, and the open-source alternatives we tested did not yield reliable progress estimates. Most notably, VITA also exceeds Meta-World’s own hand-engineered fuzzy-logic dense reward (META-WL, 0.779)—a reward function with privileged access to simulator state. That a value estimator trained exclusively on real-world video matches or exceeds this in-domain reward, despite the real-to-sim domain gap, indicates that VITA transfers to simulated reward shaping and offers practical downstream value.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the contribution of dissimilarity-based sampling by comparing sub-trajectory sampling strategies through their effect on VITA’s ability to separate expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024), which computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$, overfits to global temporal shortcuts in our setup. Random sampling injects variance by drawing sub-trajectories uniformly but does not target semantic diversity. Our dissimilarity-based sampling, which explicitly selects the sub-trajectories of maximal pairwise dissimilarity, directly promotes semantic diversity and, in consequence, outperforms both alternatives at discriminating expert from non-expert trajectories. The full analysis is deferred to Appendix F.1.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We now dissect how the mechanism for incorporating temporal context governs value function estimation, isolating the design choice at the heart of VITA. VITA performs step-by-step adaptation

486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. non-expert discrimination (BinVOC; higher is better).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on Meta-World MT10 (IQM; higher is better).

Table 2: (a) Discrimination of expert from scripted robot demonstrations, reported as average BinVOC across the five in-distribution scripted datasets; VITA matches the perfect discrimination of the autoregressive GVL variants while surpassing the recurrent CLIP-GRU and the temporally unaware CLIP baselines. (b) Offline RL on Meta-World MT10, reported as interquartile mean (IQM) return with 95% stratified bootstrap confidence intervals over 10 seeds; VITA attains the best IQM, ranking above both the CLIP-based baselines and the simulator’s fuzzy-logic dense reward (META-WL), with confidence intervals that overlap those of the learned baselines. Best result per column in **bold**.

without reset (implicit memory), incrementally updating the parameters from θ_{t-1} so that the estimator encodes the trajectory history directly into its parameters, thereby capturing both temporal order and accumulated temporal context. We contrast this with three ablated adaptation strategies that hold every other component fixed and vary only how history is aggregated: (i) *trajectory-level adaptation* (*TTT-TR*), which applies a single gradient update computed over the full trajectory; because this batched update averages the loss across all frames, it discards temporal order; (ii) *memoryless adaptation* (*TTT-RS*), which resets the adaptation module to θ_0 at every step and adapts to the current observation alone, so that no temporal information is carried forward; (iii) *explicit memory* (*TTT-EX*), which likewise resets the module at every step but updates over a local window of recent frames matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA consistently and substantially outperforms all three variants, establishing that it is the incremental, per-timestep *accumulation* of temporal context—not the presence of temporal information per se—that drives accurate value function estimation. The complete analysis is provided in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been employed as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), relying on frame-level similarity scores that afford no temporal modeling. A separate line fine-tunes CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to produce dense rewards, but the requisite supervision curtails zero-shot applicability. RoboCLIP (Sontakke et al., 2023) enables one-shot reward specification without fine-tuning, yet yields only sparse rewards and still depends on demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions that lean on scale for generalization, whereas ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding, combined with a handful of in-domain demonstrations. VITA departs from all of these by adapting a pre-trained contrastive vision–language encoder *at inference*, capturing temporal and semantic context through test-time adaptation and thereby dispensing with both large-scale multimodal pre-training and additional domain-specific demonstrations. Because it is agnostic to the underlying encoder, VITA applies to any pre-trained vision–language representation, augmenting its generalization and temporal reasoning through meta-learned test-time adaptation. Autoregressive VLMS (Alayrac et al., 2022) exploit in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) counteracts the monotonic bias induced by ordered pre-training data

by shuffling frames at inference, but this remedy sacrifices temporal order—the very signal needed for temporal reasoning and for disambiguating visually similar yet temporally distinct states. VITA instead preserves chronological order while capturing temporal context through test-time adaptation, sidestepping the trade-off altogether.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers parameter-efficient means of adapting VLMs without full fine-tuning. Prompt-based methods optimize parameters at test time from the task context, although their underlying mechanisms remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argued that CLIP representations are amenable to test-time adaptation, motivated by the finding that CLIP encodes human-interpretable concepts surfaced through mechanistic interpretability (Bricken et al., 2023), and CLIP-based similarity has itself been used as a reward whose parameters are optimized at test time via reinforcement learning (Zhao et al., 2024). Our work builds instead on test-time training (Sun et al., 2020), in which an adaptation module is updated at each timestep by a self-supervised loss. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023) that require task-specific adaptation episodes during training, test-time training permits direct online adaptation at inference without task labels—a property VITA exploits to attain zero-shot generalization across manipulation tasks, environments, and embodiments.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our experiments establish that test-time adaptation equips value functions for robotic manipulation with robust generalization across distribution shifts in task, environment, and embodiment. Beyond generalization, VITA reliably distinguishes expert from non-expert visual trajectories and furnishes an effective zero-shot reward-shaping signal, opening downstream applications in RL and imitation learning. A single mechanism underlies all three results—sequential per-timestep updating: by folding the observation history into its parameters, VITA captures temporal and semantic context more effectively than either the CLIP-based baselines or in-context learning with autoregressive VLMs. That VITA outperforms CLIP-GRU on both expert–non-expert discrimination and offline RL indicates that implicit memory formed through sequential test-time updates generalizes better than temporal history stored explicitly in recurrent hidden states. Pre-trained autoregressive VLMs such as GVL, for all their breadth of generalization, are not trained for progress estimation—a task that demands temporal reasoning over visual trajectories—and are correspondingly outperformed. Finally, because only a lightweight adaptation module is updated at test time, the per-timestep adaptation overhead is negligible, so VITA preserves real-time applicability.

6.2 LIMITATIONS AND FUTURE WORK

While test-time adaptation improves the generalization of zero-shot progress estimation across the unseen tasks and environments in our experiments, our evaluation remains centered on short-horizon robotic manipulation, and settings with high execution variability or substantially extended horizons can still pose challenges; because our progress metric is grounded in monotonic expert time, trajectories that backtrack, retry, or admit multiple valid strategies fall outside its present scope. Moreover, although the adaptation overhead is negligible owing to the lightweight adaptation module, updating a value function estimator at every timestep may be unsafe during deployment, which could constrain applicability in some real-time scenarios. These considerations motivate several directions for future work. We plan to investigate alternative test-time adaptation schemes for vision–language models, particularly for training agents within world models, and to extend VITA to real-time closed-loop control and more complex RL environments. Because VITA is agnostic to the underlying vision–language encoder, it is a natural mechanism for enhancing the generalization and temporal reasoning of any domain-specific pre-trained multimodal representation through meta-learned test-time adaptation. Finally, although we provide empirical evidence that dissimilarity-based sampling mitigates shortcut learning, a theoretical account of how diversity-based sampling shapes shortcut learning remains open; this lies beyond the scope of our core contribution,

which is a value function estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

REFERENCES

- Niket Agarwal, Arslan Ali, Maciej Bala, Yogesh Balaji, Erik Barker, Tiffany Cai, Prithvijit Chattopadhyay, Yongxin Chen, Yin Cui, Yifan Ding, et al. Cosmos world foundation model platform for physical ai. *arXiv preprint arXiv:2501.03575*, 2025.
- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, 2021.
- Minttu Alakuijala, Reginald McLean, Isaac Woungang, Nariman Farsad, Samuel Kaski, Pekka Marttinen, and Kai Yuan. Video-language critic: Transferable reward functions for language-conditioned robotics. *arXiv preprint arXiv:2405.19988*, 2024.
- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andy Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. *arXiv preprint arXiv:2204.14198*, 2022.
- Bowen Baker, Ilge Akkaya, Peter Zhokov, Joost Huizinga, Jie Tang, Adrien Ecoffet, Brandon Houghton, Raul Sampedro, and Jeff Clune. Video pretraining (vpt): Learning to act by watching unlabeled online videos. In *Advances in Neural Information Processing Systems*, 2022.
- Jakob Bauer, Kate Baumli, Feryal M. P. Behbahani, Avishkar Bhoopchand, Nathalie Bradley-Schmieg, Michael Chang, Natalie Clay, Adrian Collister, Vibhavari Dasagi, Lucy Gonzalez, Karol Gregor, Edward Hughes, Sheleem Kashem, Maria Loks-Thompson, Hannah Openshaw, Jack Parker-Holder, Shreya Pathak, Nicolas Perez Nieves, Nemanja Rakicevic, Tim Rocktäschel, Yannick Schroecker, Satinder Singh, Jakub Sygnowski, Karl Tuyls, Sarah York, Alexander Zacherl, and Lei M. Zhang. Human-timescale adaptation in an open-ended task space. In *International Conference on Machine Learning*, 2023.
- Kate Baumli, Satinder Baveja, Feryal M. P. Behbahani, Harris Chan, Gheorghe Comanici, Sebastian Flennerhag, Maxime Gazeau, Kristian Holsheimer, Dan Horgan, Michael Laskin, Clare Lyle, Hussain Masoom, Kay McKinney, Volodymyr Mnih, Alexander Neitz, Fabio Pardo, Jack Parker-Holder, John Quan, Tim Rocktäschel, Himanshu Sahni, Tom Schaul, Yannick Schroecker, Stephen Spencer, Richie Steigerwald, Luyu Wang, and Lei Zhang. Vision-language models as a source of rewards. *arXiv preprint arXiv:2312.09187*, 2023.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Krzysztof Choromanski, Tianli Ding, Danny Driess, Kumar Avinava Dubey, Chelsea Finn, Peter R. Florence, Chuyuan Fu, Montse Gonzalez Arenas, Keerthana Gopalakrishnan, Kehang Han, Karol Hausman, Alexander Herzog, Jasmine Hsu, Brian Ichter, Alex Irpan, Nikhil J. Joshi, Ryan C. Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Sergey Levine, Henryk Michalewski, Igor Mordatch, Karl Pertsch, Kanishka Rao, Krista Reymann, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi, Pierre Sermanet, Jaspiar Singh, Anikait Singh, Radu Soricut, Huong Tran, Vincent Vanhoucke, Quan Ho Vuong, Ayzaan Wahid, Stefan Welker, Paul Wohlhart, Ted Xiao, Tianhe Yu, and Brianna Zitkovich. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, 2023.

648 Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes,
649 Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Maria Elis-
650 abeth Bechtle, Feryal Behbahani, Stephanie C.Y. Chan, Nicolas Heess, Lucy Gonzalez, Simon
651 Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando De Freitas,
652 Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. In *International
653 Conference on Machine Learning*, 2024.

654 Annie S Chen, Suraj Nair, and Chelsea Finn. Learning generalizable robotic reward functions from
655 in-the-wild human videos. *arXiv preprint arXiv:2103.16817*, 2021.

656 Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger
657 Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for
658 statistical machine translation. *arXiv preprint arXiv:1406.1078*, 2014.

659 Nitish Dashora, Dibya Ghosh, and Sergey Levine. Viva: Video-trained value functions for guiding
660 online rl from diverse data. *arXiv preprint arXiv:2503.18210*, 2025.

661 Yuqing Du, Ksenia Konyushkova, Misha Denil, Akhil Raju, Jessica Landon, Felix Hill, Nando
662 de Freitas, and Serkan Cabi. Vision-language models as success detectors. In *Conference on
663 Lifelong Learning Agents*, 2023.

664 Ashley Edwards, Himanshu Sahni, Yannick Schroecker, and Charles Isbell. Imitating latent policies
665 from observation. In *International Conference on Machine Learning*, 2019.

666 Lasse Espeholt, Hubert Soyer, Remi Munos, Karen Simonyan, Vlad Mnih, Tom Ward, Yotam
667 Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA:
668 Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International
669 Conference on Machine Learning*, 2018.

670 Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, Haoyi Zhu, Andrew Tang,
671 De-An Huang, Yuke Zhu, and Anima Anandkumar. Minedojo: Building open-ended embodied
672 agents with internet-scale knowledge. In *Advances in Neural Information Processing Systems*,
673 2022.

674 Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of
675 deep networks. In *International Conference on Machine Learning*, 2017.

676 Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias
677 Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine
678 Intelligence*, 2020.

679 Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett
680 Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal
681 understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.

682 Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT Press, 2016.

683 David Ha and Jürgen Schmidhuber. World models. In *Advances in Neural Information Processing
684 Systems*, 2018.

685 Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering atari with discrete
686 world models. In *International Conference on Learning Representations*, 2021.

687 Jonathan Ho and Stefano Ermon. Generative adversarial imitation learning. In *Advances in Neural
688 Information Processing Systems*, 2016.

689 Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural computation*, 1997.

690 Anthony Hu, Lloyd Russell, Hudson Yeo, Zak Murez, George Fedoseev, Alex Kendall, Jamie Shotton,
691 and Gianluca Corrado. GAIA-1: A generative world model for autonomous driving. In *Proceedings
692 of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

693 Edward Hughes, Michael D. Dennis, Jack Parker-Holder, Feryal M. P. Behbahani, Aditi Mavalankar,
694 Yuge Shi, Tom Schaul, and Tim Rocktäschel. Position: Open-endedness is essential for artificial
695 superhuman intelligence. In *International Conference on Machine Learning*, 2024.

-
- Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- Haobin Jiang, Junpeng Yue, Hao Luo, Ziluo Ding, and Zongqing Lu. Reinforcement learning friendly vision-language model for minecraft. In *European Conference on Computer Vision*, 2024.
- Muhammad Uzair Khattak, Hanoona Rasheed, Muhammad Maaz, Salman Khan, and Fahad Shahbaz Khan. Maple: Multi-modal prompt learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Paul Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model. In *Conference on Robot Learning*, 2024.
- Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning. *arXiv preprint arXiv:2110.06169*, 2021.
- Youngwoon Lee, Andrew Szot, Shao-Hua Sun, and Joseph J. Lim. Generalizable imitation learning from observation via inferring goal proximity. In *Advances in Neural Information Processing Systems*, 2021a.
- Youngwoon Lee, Andrew Szot, Shao-Hua Sun, and Joseph J. Lim. Generalizable imitation learning from observation via inferring goal proximity. In *Advances in Neural Information Processing Systems*, 2021b.
- Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. Offline reinforcement learning: Tutorial, review, and perspectives on open problems. *arXiv preprint arXiv:2005.01643*, 2020.
- Jianxiong Li, Jinliang Zheng, Yinan Zheng, Liyuan Mao, Xiao Hu, Si Cheng, Haoyi Niu, Jihao Liu, Yu Liu, Jingjing Liu, Ya-Qin Zhang, and Xianyu Zhan. Decisionnnc: Embodied multimodal representations via implicit preference learning. In *International Conference on Machine Learning*, 2024.
- Hyesu Lim, Jinho Choi, Jaegul Choo, and Steffen Schneider. Sparse autoencoders reveal selective remapping of visual concepts during adaptation. In *International Conference on Learning Representations*, 2025.
- Ziyi Lin, Honglie Liu, Andrew Zisserman, and Andrea Vedaldi. Frozen clip models are efficient video learners. In *European Conference on Computer Vision*, 2022.
- Xudong Ma, Chen Liang, Zhecan Wang, Chuang Gan, and Zhiting Li. Swapprompt: Test-time prompt adaptation for vision-language models. In *Advances in Neural Information Processing Systems*, 2023a.
- Yecheng Jason Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy Zhang. Vip: Towards universal visual reward and representation via value-implicit pre-training. *arXiv preprint arXiv:2210.00030*, 2022.
- Yecheng Jason Ma, Vikash Kumar, Amy Zhang, Osbert Bastani, and Dinesh Jayaraman. LIV: Language-image representations and rewards for robotic control. In *International Conference on Machine Learning*, 2023b.
- Yecheng Jason Ma, Joey Hejna, Ayzaan Wahid, Chuyuan Fu, Dhruv Shah, Jacky Liang, Zhuo Xu, Sean Kirmani, Peng Xu, Danny Driess, Ted Xiao, Jonathan Tompson, Osbert Bastani, Dinesh Jayaraman, Wenhao Yu, Tingnan Zhang, Dorsa Sadigh, and Fei Xia. Vision language models are in-context value learners. In *International Conference on Learning Representations*, 2024.
- Yutaka Matsuo, Yann LeCun, Maneesh Sahani, Doina Precup, David Silver, Masashi Sugiyama, Eiji Uchibe, and Jun Morimoto. Deep learning, reinforcement learning, and world models. *Neural Networks*, 2022.

- Viorica Pătrăucean, Yusuf Aytar, Roozbeh Mottaghi, Oleh Rybkin, Mario Lucic, Carl Vondrick, William T. Freeman, and Andrew Zisserman. Perception test: A diagnostic benchmark for multimodal video models. In *Advances in Neural Information Processing Systems*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.
- Scott Reed, Konrad Zolna, Emilio Parisotto, Sergio Gomez Colmenarejo, Alexander Novikov, Gabriel Barth-Maron, Mai Giménez, Yuri Sulsky, Jackie Kay, Jost Tobias Springenberg, Tom Eccles, Jake Bruce, Ali Razavi, Ashley D. Edwards, Nicolas Manfred Otto Heess, Yutian Chen, Raia Hadsell, Oriol Vinyals, Mahyar Bordbar, and Nando de Freitas. A generalist agent. *arXiv preprint arXiv:2205.06175*, 2022.
- Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations*, 2024.
- Dominik Schmidt and Minqi Jiang. Learning to act without actions. In *International Conference on Learning Representations*, 2024.
- David Silver and Richard S. Sutton. Welcome to the era of experience. Technical report, Google AI, 2025.
- SIMA Team, Maria Abi Raad, Arun Ahuja, Catarina Barros, Frederic Besse, Andrew Bolt, Adrian Bolton, Bethanie Brownfield, Gavin Buttimore, Max Cant, et al. Scaling instructable agents across many simulated worlds. *arXiv preprint arXiv:2404.10179*, 2024.
- Sumedh Sontakke, Jesse Zhang, Sébastien M. R. Arnold, Karl Pertsch, Erdem Biyik, Dorsa Sadigh, Chelsea Finn, and Laurent Itti. Roboclip: One demonstration is enough to learn robot policies. In *Advances in Neural Information Processing Systems*, 2023.
- Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2020.
- Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): Rnns with expressive hidden states. *arXiv preprint arXiv:2407.04620*, 2024.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.
- Homer Rich Walke, Kevin Black, Tony Z. Zhao, Quan Vuong, Chongyi Zheng, Philippe Hansen-Estruch, Andre Wang He, Vivek Myers, Moo Jin Kim, Max Du, Abraham Lee, Kuan Fang, Chelsea Finn, and Sergey Levine. Bridgedata v2: A dataset for robot learning at scale. In *Conference on Robot Learning*, 2023.
- Limin Wang, Yuanjun Xiong, Dahua Lin, and Luc Van Gool. Temporal segment networks: Towards good practices for deep action recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2018.
- Renhao Wang, Yu Sun, Yossi Gandelsman, Xinlei Chen, Alexei A. Efros, and Xiaolong Wang. Test-time training on video streams. *arXiv preprint arXiv:2307.05014*, 2023.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

-
- Ze Yang, Yiming Zhang, Jerry Yu, Junru Cai, and Ming-Hsuan Yang. UniSim: A neural closed-loop sensor simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning*, 2020.
- Chuheng Zhang, Tim Pearce, Pushi Zhang, Kaixin Wang, Xiaoyu Chen, Wei Shen, Li Zhao, and Jiang Bian. What do latent action models actually learn? *arXiv preprint arXiv:2506.15691*, 2025a.
- Jiahui Zhang, Yusen Luo, Abrar Anwar, Sumedh Anand Sontakke, Joseph J Lim, Jesse Thomason, Erdem Biyik, and Jesse Zhang. RewiND: Language-guided rewards teach robot policies without new demonstrations. In *Conference on Robot Learning*, 2025b.
- Shuyuan Zhao, Xin Wang, Linchao Zhu, and Yi Yang. Test-time adaptation with clip reward for zero-shot generalization in vision-language models. *arXiv preprint arXiv:2401.12345*, 2024.
- Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 2022.

A TRAINING DATASET

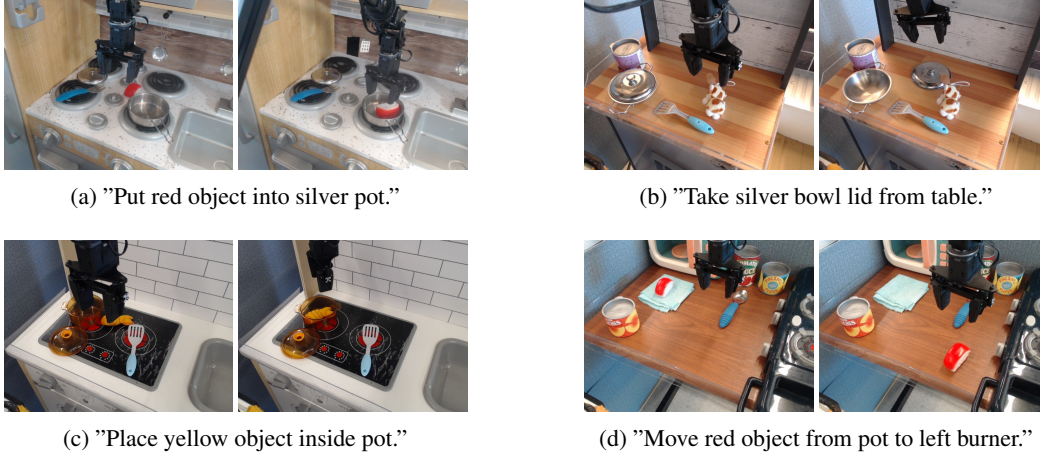


Figure 4: Each subfigure shows the start and end frames of an expert demonstration used for training, together with its natural-language task description. All demonstrations are pick-and-place tasks collected across four distinct ToyKitchen configurations with the WidowX 250 embodiment, defining the deliberately narrow training distribution from which VITA must generalize.

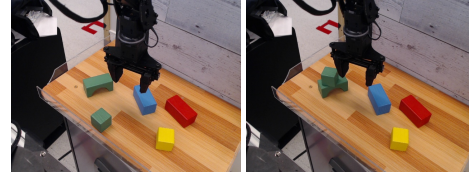
B EVALUATION DATASETS

Table 3: Evaluation datasets, characterized by task type, environment, and embodiment. The datasets span pick-and-place, folding, sweeping, and stacking tasks across five environments and two embodiments (WidowX 250 and DeepThought), thereby probing generalization independently along each of the three shift axes and under their combination.

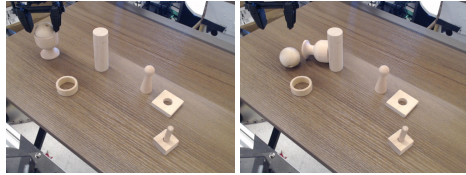
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



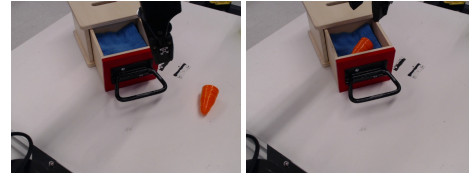
(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

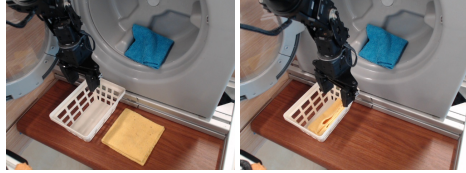


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure shows the start and end frames of an evaluation trajectory under embodiment shift, together with its natural-language task description. The top row holds the environment fixed (ToyKitchen) while switching to the DeepThought robot, thereby isolating embodiment shift in isolation; the bottom row compounds this with new environments, jointly stressing embodiment and environment generalization.



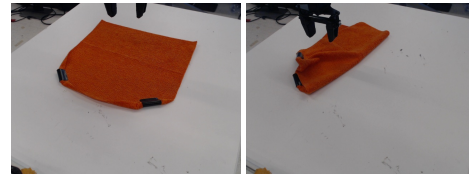
(a) lm_pnp: "Place blue cloth inside washer."



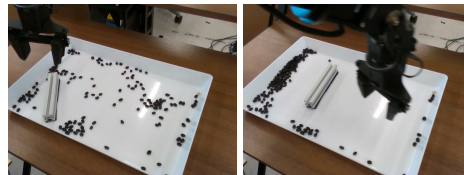
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure shows the start and end frames of an evaluation trajectory under environment shift, together with its natural-language task description. The embodiment (WidowX 250) is held fixed throughout, so that variation arises solely from the visually and structurally distinct environments, thereby isolating environment generalization.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure shows the start and end frames of a scripted evaluation trajectory collected in the ToyKitchen environment with the WidowX 250 robot. All trajectories are pick-and-place behaviors spanning varied object categories, generated by a randomized controller and therefore departing from the expert-like monotonic progress on which the expert demonstrations rest. Generic task descriptions are used, as per-trajectory annotations are not publicly available.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We adopt the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone, encoding each visual observation and task description with the CLIP vision and text encoders, respectively. The joint representation is formed by concatenation rather than element-wise product, a choice motivated by superior validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activations and projection dimension $d' = 64$. Optimization uses AdamW with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine schedule with 10% warmup; the batch size is 32, and all trajectories are padded to a maximum length of 120 frames, matching the longest training sequence. The self-supervised weighting coefficient λ_{self} is set to 0.5, selected on the validation set. We train for 5 epochs, since validation VOC plateaus early and extended training yields no further gains. Unless otherwise specified, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, we adapt only the temporal adaptation module f_{adapt} , retaining the training-time projection dimension ($d' = 64$), through a single gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. This configuration was selected from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$

Table 4: Sensitivity of VITA to the test-time learning rate η and the number of adaptation steps t_{ep} , measured by validation VOC. Best result per column in **bold**; the single-step, large- η setting ($\eta = 0.1$, $t_{\text{ep}} = 1$) attains the overall best score while incurring the lowest inference cost.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

and adaptation steps $\{1, 5, 10\}$ on the basis of the validation results in Table 4. Two regimes prove comparably effective—(i) a single update with a larger learning rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and (ii) multiple updates with a smaller learning rate ($\eta = 0.01$, $t_{\text{ep}} = 10$)—and we adopt the former, which is marginally more accurate while requiring the fewest adaptation steps and hence the lowest inference overhead. The two-way trade-off underscores that VITA’s benefit derives from the sequential structure of the updates rather than from their sheer number.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, selected through the hyperparameter tuning described above. CLIP-FT inherits VITA’s architecture but omits the adaptation module and the self-supervised loss: it applies a frozen CLIP encoder followed by a linear projection and a two-layer MLP to predict task progress, trained by standard supervised regression without meta-learning or test-time adaptation. So that the comparison does not disadvantage this baseline on capacity, we grant it an $8\times$ larger projection matrix and $10\times$ more training steps than VITA (Lin et al., 2022). For VLM-RM we use a baseline prompt describing the environment. CLIP-GRU likewise inherits VITA’s frozen CLIP encoder and regression head h , but replaces the test-time adaptation module f_{adapt} with a Gated Recurrent Unit (GRU) that encodes temporal history explicitly through recurrent hidden states; the GRU has a single layer with hidden size 64, matching VITA’s projection dimension d' , and its final hidden state is fed to h , so that the two temporal mechanisms are compared at a comparable capacity while sharing an identical encoder and regression head. For both GVL-0S and GVL-1S we use the latest Gemini 1.5 Pro release, `gemini-1.5-pro-latest` (Gemini Team et al., 2024), an upgrade over the earlier `gemini-1.5-pro` used in the original GVL implementation. We additionally evaluated GVL with the open-source autoregressive VLM `Qwen-VL 2.5` (Yang et al., 2024), which failed to overcome the monotonic temporal bias despite frame shuffling, whereas GPT-4o (Hurst et al., 2024) consistently declined to emit scalar progress estimates in our setup. All GVL evaluations followed the prompt template introduced in the original work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We employ Implicit Q-Learning (IQL) with a single configuration held fixed across all tasks, so that any performance difference is attributable to the reward signal rather than to per-task tuning. Expectile regression uses $\tau = 0.7$, with advantage-weighting temperature 3.0 and clipping threshold 100.0. The actor and critic use learning rates 1×10^{-4} and 3×10^{-4} , respectively, at batch size 256. Policies are trained for 100,000 gradient steps and evaluated every 10,000 steps on 20 rollouts per task (horizon 150). The implementation follows `d3rlpy`, with configuration: `expectile=0.7`, `weight_temp=3.0`, `max_weight=100.0`, `actor_lr=1e-4`, `critic_lr=3e-4`, `batch_size=256`.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. The IQM averages the middle 50% of scores and is therefore markedly less sensitive to outliers than the mean or median. To quantify uncertainty, we estimate 95% confidence intervals via stratified bootstrap with 10,000 resamples, stratified by task. This protocol is applied uniformly to every method in the Meta-World MT10 experiments, so that the reported differences are accompanied by an explicit characterization of their statistical uncertainty.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC across the scripted datasets under different training-time sampling strategies for test-time adaptation; a checkmark denotes successful discrimination between expert and suboptimal trajectories. The strategies **FT** (full-trajectory), **Rand** (random), and **Diss** (dissimilarity-based) are parameterized by (w_{tr}, k) . Dissimilarity-based sampling with (8, 8) achieves perfect discrimination on all five datasets, whereas full-trajectory sampling succeeds on only one.

Dataset	FT (8,–)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc.pnp_obj		✓	✓		✓
sc.pnp_robj	✓	✓	✓	✓	✓
sc.pnp_stoy			✓		✓
sc.pnp_uten	✓	✓	✓	✓	✓
sc.pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We validate the proposed dissimilarity-based sampling by quantifying how the training-time sub-trajectory sampling strategy shapes the model’s ability to discriminate expert from suboptimal trajectories. Table 5 reports Disc@VOC over the five scripted datasets, recording whether a model assigns a higher average VOC to expert trajectories (`tk_pnp`) than to each evaluation dataset. We contrast three strategies: (1) full-trajectory sampling, which adapts over all overlapping sub-trajectories of length w_{tr} with stride s , following prior work (Sun et al., 2024); (2) random sampling, which draws k sub-trajectories of length w_{tr} uniformly per video; and (3) our dissimilarity-based sampling, which builds a candidate set of sub-trajectories of length w_{tr} with stride s and retains the k sub-trajectories that maximize pairwise dissimilarity in the representation space.

Dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ attains perfect discriminative performance (Disc@VOC = 1.0). Selecting fewer windows ($k = 4$) at the same length $w_{tr} = 8$ matches it, whereas shortening the sub-trajectory to $w_{tr} = 4$ degrades effectiveness (0.60). The full-trajectory baseline is the weakest (0.20), confirming that adapting over all consecutive windows overfits to global temporal cues; random sampling improves on it (0.8) yet remains less reliable than dissimilarity-based selection. Taken together, these results demonstrate that both sub-trajectory diversity and semantic locality are necessary for discriminative test-time adaptation, and that dissimilarity-based sampling is the mechanism that secures both.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We evaluate four adaptation strategies that differ solely in how temporal context is incorporated: (1) TTT-TR, which applies a single gradient update computed over the full trajectory (offline); this batched update averages the loss across all frames and thereby discards temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates from the current observation alone (memoryless), so that no temporal information is carried forward; (3) TTT-EX, which likewise resets the module at every step but updates over a local window of recent frames matching the meta-training window size ($w_{tr} = 8$); averaging gradients over this window discards the temporal ordering within it; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory), forming an implicit memory through the sequential accumulation of temporal history in the

Table 6: Validation VOC for progress estimation under distribution shift, comparing the three ablated adaptation strategies against VITA; best per row in **bold**. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift. VITA attains the best VOC on every dataset by a wide margin, while the three batched-update variants remain close to the memoryless baseline throughout.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

adapter parameters. To keep the comparison fair, we ran identical test-time hyperparameter sweeps for all variants, as described in Appendix D: VITA, TTT-RS, and TTT-TR are optimal at learning rate 0.1 with a single adaptation step, whereas TTT-EX performs best at learning rate 1.0 with one step.

Table 6 reveals a sharp separation. TTT-TR, TTT-RS, and TTT-EX perform comparably, showing that batched updates—whether over a short explicit window (TTT-EX) or over the full trajectory (TTT-TR)—are essentially no better than the memoryless baseline (TTT-RS). VITA, by contrast, outperforms all three by a wide margin on every dataset—for instance, 0.7822 versus at most 0.1957 in-distribution—establishing that sequential, per-timestep updates are the decisive ingredient for accurate value function estimation in real-world robotic manipulation. This behavior follows directly from the structure of implicit memory: updating over a local window would double-count the most recent observations—once explicitly through the window and once implicitly through the accumulated parameters—whereas updating less frequently would omit intermediate frames and forfeit information. The implicit-memory test-time training paradigm therefore requires precisely the sequential, per-timestep updates without reset that VITA performs.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse subset selection while avoiding the exponential cost of enumerating all $\binom{N_c}{b}$ subsets in Eq. 4, we adopt a scoring-based heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others—equivalently, the row sum of the distance matrix. With $N_c = T - w_{tr} + 1$ candidate windows, each represented by a flattened feature vector of dimension $m = w_{tr} \cdot d$ (where d is the dimension of each multimodal representation z_t), we compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU through batched `cdist` at a cost of $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ operations.

In our experiments, training trajectories are short (mean length 69, maximum length 120). With batch size 32, feature dimension $d = 1024$ (the CLIP-based multimodal representation), and window length $w_{tr} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to the cost of a single forward and backward pass through our model, confirming that dissimilarity-based sampling secures diverse-window selection at essentially no computational expense.